ECO-GUARD

Intelligent Economic Inspector

An AI-Powered Multi-Agent System for
Household Survey Data Verification

TEAM — GROUPE 02

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SW 413 · 2nd Semester –2026

Project Idea & Objectives

THE OPPORTUNITY

Statistical authorities conduct large-scale household surveys used in national planning and policy decisions.

To elevate data quality, Eco-Guard aims to:

- Automate data verification at scale
- Enhance accuracy through AI analysis
- Accelerate quality control workflows
- Replace manual review with intelligent pipeline

OUR SOLUTION

Three independent AI agents working in parallel:

Demographic Agent

Checks age, education & occupation consistency

Financial Agent

Compares income/expenses vs regional benchmarks

Manager Agent

Synthesizes final trust score (0–100) + recommendation

System Architecture

Decoupled Architecture

Frontend and Backend fully separated

User Browser

eco-guard.streamlit.app

↓

Streamlit Cloud — Frontend

app.py · api_client.py · database.py

HTTP REST Calls

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Railway FastAPI — Backend

/agent/demographic · /agent/financial · /agent/manager

Claude API Calls

↓

Anthropic Claude Sonnet

AI Analysis Engine — Real Claude API calls per agent

Analysis Pipeline — How It Works

01

Input

Manual entry or CSV batch upload

›

02

Preprocess

Data cleaned & prepared for AI

›

03

AI Agents

3 agents run in parallel

›

04

Trust Score

Manager synthesizes 0–100 score

›

05

Visualize

Charts + export to Excel

Dashboard Features

RTL Arabic Interface

Full Arabic right-to-left layout with Tajawal font and glassmorphism design

Role-Based Auth

Field Researcher & Admin roles with PBKDF2 password hashing

Manual Analysis

Enter individual records and get instant trust score with 3-agent breakdown

Batch CSV Processing

Upload datasets with parallel processing and live progress tracking

Interactive Charts

Plotly gauge chart for trust score + bar chart comparing all agents

Excel Export

Color-coded report: green (trusted), orange (review), red (rejected)

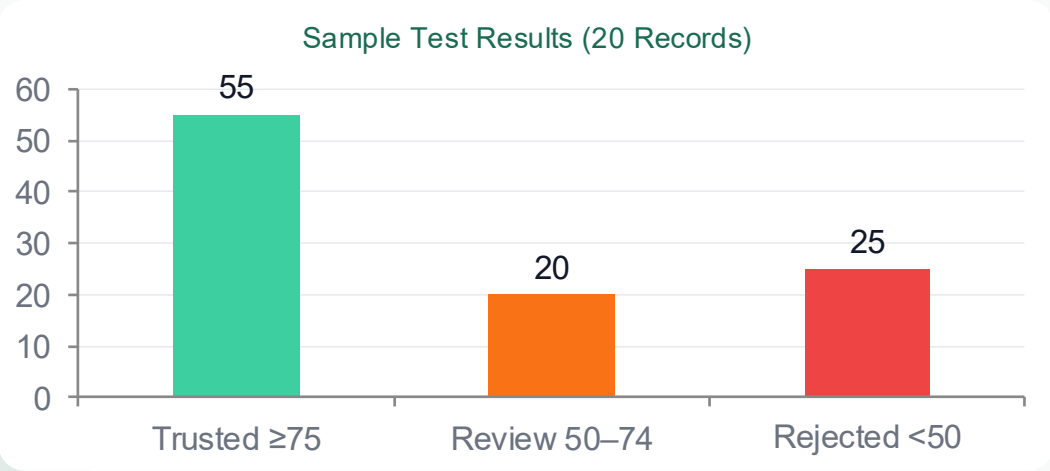
Results & Performance

3
AI Agents
Independent endpoints

~60%
Faster
Parallel processing

10
Data Fields
Per record

Live
Deployed
Streamlit + Railway



Agents Successfully Detected:

- Age–education inconsistencies
- Income 99% below regional median
- Expense/income ratio >90%
- Electricity bills vs housing mismatch

Technology Stack & Deployment

Frontend	Python · Streamlit · Plotly · CSS (RTL / Glassmorphism)
Backend	FastAPI · Uvicorn · Python 3.12 · REST API
AI Engine	Anthropic Claude Sonnet · Multi-Agent System · JSON Parsing
Data	SQLite · PBKDF2 Auth · GASTAT Reference Data (JSON)
DevOps	Railway (Backend) · Streamlit Cloud (Frontend) · GitHub CI/CD
Libraries	pandas · openpyxl · concurrent.futures · requests · python-dotenv

Thank You

ECO-GUARD · Intelligent Economic Inspector

Live App eco-guard.streamlit.app

GitHub github.com/Haneen-Alshuwayqi/Eco-Guard

Eco-Guard delivers scalable, AI-powered data verification — fully deployed and ready for real-world use.